

# Software Development Blueprint: SkillSwap

**Status:** Draft

**Version:** 0.1

**Blueprint ID:** BP-001

**Owner:** Group 4

**Last updated:** 2026-09-15

## 1. Executive Summary

- **Business problem:** Sinh viên có kỹ năng đa dạng nhưng thiếu thời gian và ngân sách. Họ muốn học kỹ năng mới (lập trình, thiết kế, ngôn ngữ...) nhưng không đủ khả năng chi trả cho các khóa học chính quy, và không có cách đáng tin cậy để tìm và trả tiền cho một sinh viên khác thực sự biết về lĩnh vực đó. Đồng thời, sinh viên có kỹ năng không có kênh dễ dàng để dạy và kiếm tiền từ những gì họ biết.
- **Desired outcome:** Nền tảng kết nối sinh viên dạy/học kỹ năng với ví điện tử nội bộ. Giáo viên được xác minh kỹ năng/chứng chỉ trước khi mở lớp. Học viên nạp tiền thật → credit → mua lớp học. Giáo viên nhận credit → rút tiền thật. Platform lấy 10% commission.
- **Success metrics:**
  - Tỷ lệ học viên hoàn thành ít nhất 1 lớp học
  - Số lượng giáo viên đã xác minh hoạt động
  - Thời gian trung bình từ đăng ký đến lớp học đầu tiên
  - Rating trung bình của giáo viên
  - Tỷ lệ dispute (khiếu nại) trên tổng giao dịch
- **Recommended direction:** Mobile-first web responsive, MVP tập trung core wallet + booking flow, tích hợp Jitsi cho video call, manual verification cho đến khi có đủ data feedback.

## 2. Scope

### In scope

#### 1. Authentication & Authorization

- Đăng ký, đăng nhập (email/password)
- Student verification (upload student ID/enrollment confirmation)
- Teacher verification (submit certificates + verifier approval)

#### 2. Digital Wallet System

- Nạp tiền qua payment gateway → credit
- Debit khi học viên book lớp

- Credit cho giáo viên khi lớp hoàn thành
- Rút credit → tiền thật
- 10% commission trên mỗi giao dịch
- Transaction history

### 3. Skill & Class Management

- Skill categories (do system admin quản lý)
- Teacher tạo class với thông tin: title, description, skill, price, duration (30min-3h), schedule
- Booking với rule: book ít nhất 24h trước
- Class states: scheduled, in\_progress, completed, cancelled

### 4. Video Call Integration

- Jitsi integration cho online classes

### 5. Messaging System

- In-platform chat giữa learner và teacher

### 6. Rating & Review System

- Post-class reviews với rating + comment
- Low-rated profiles giảm visibility

## Out of scope

- Native mobile app (iOS/Android)
- AI-based skill matching/recommendation
- Automatic refunds
- Integration với school database
- Multi-language (chỉ Vietnamese/English)
- Dedicated report/moderation system
- Auto student verification qua school email

## 3. Stakeholders and Users

| ID     | Role    | Responsibility                      | Decision/approval             |
|--------|---------|-------------------------------------|-------------------------------|
| ST-001 | Learner | Học sinh/sinh viên muốn học kỹ năng | Chọn lớp, book, đánh giá      |
| ST-002 | Teacher | Sinh viên đã xác minh, mở lớp dạy   | Tạo lớp, set giá, nhận credit |

|        |                 |                                                           |                                |
|--------|-----------------|-----------------------------------------------------------|--------------------------------|
| ST-003 | Verifier        | Chuyên gia domain, xác minh skill/certificate của teacher | Phê duyệt/từ chối verification |
| ST-004 | Admin           | Quản lý skill categories, user issues                     | Điều phối hệ thống             |
| ST-005 | Payment Gateway | Xử lý real-money transactions                             | Out of scope (chọn sau)        |

## 4. Assumptions, Constraints and Open Questions

| ID    | Type          | Statement                                                     | Owner   | Status      |
|-------|---------------|---------------------------------------------------------------|---------|-------------|
| A-001 | Assumption    | Sinh viên có tài khoản ngân hàng hoặc e-wallet (MoMo/ZaloPay) | Group 4 | Unconfirmed |
| A-002 | Assumption    | Sinh viên sẵn sàng upload ảnh student ID                      | Group 4 | Unconfirmed |
| A-003 | Assumption    | Có đủ verifiers có domain expertise trước launch              | Group 4 | Unconfirmed |
| A-004 | Assumption    | Sinh viên có internet ổn định cho video call                  | Group 4 | Unconfirmed |
| C-001 | Constraint    | Chỉ online classes, không offline                             | Intent  | Confirmed   |
| C-002 | Constraint    | 10% commission trên mỗi giao dịch                             | Intent  | Confirmed   |
| C-003 | Constraint    | Session duration: 30 min - 3 hours                            | Intent  | Confirmed   |
| C-004 | Constraint    | Book ít nhất 24h trước                                        | Intent  | Confirmed   |
| C-005 | Constraint    | 1 credit = 1,000 VND                                          | Intent  | Confirmed   |
| C-006 | Constraint    | Mobile-first design                                           | Intent  | Confirmed   |
| O-001 | Open Question | Cấu trúc skill categories (hierarchical hay flat?)            | Group 4 | Open        |

|       |               |                                        |         |           |
|-------|---------------|----------------------------------------|---------|-----------|
| O-002 | Open Question | Threshold rating để ẩn profile?        | Group 4 | Open      |
| O-003 | Open Question | Refund policy cho class không diễn ra? | Group 4 | Open      |
| D-001 | Decision      | Teachers tự đặt giá class              | User    | Confirmed |
| D-002 | Decision      | Launch với 1 trường trước              | User    | Confirmed |

## 5. Process Analysis

### 5.1 Learner Flow

| Step | Current state | Future state                           | Actor/system                       | Rule or exception              |
|------|---------------|----------------------------------------|------------------------------------|--------------------------------|
| 1    | N/A           | Learner đăng ký + verify student ID    | Learner → System                   | Admin review trong 24-48h      |
| 2    | N/A           | Learner nạp tiền → credit              | Learner → Payment Gateway → Wallet | Min top-up amount TBD          |
| 3    | N/A           | Learner browse/search skills & classes | Learner → System                   | Filter by skill, price, rating |
| 4    | N/A           | Learner book class (trả credit)        | Learner → System                   | Phải book 24h trước, đủ credit |
| 5    | N/A           | Learner join video call                | Learner → Jitsi                    | Link được gửi khi đến giờ      |
| 6    | N/A           | Learner đánh giá sau class             | Learner → System                   | Rating 1-5 + comment           |

### 5.2 Teacher Flow

| Step | Current state | Future state                                 | Actor/system       | Rule or exception       |
|------|---------------|----------------------------------------------|--------------------|-------------------------|
| 1    | N/A           | Teacher đăng ký + verify student ID          | Teacher → System   | Same as learner         |
| 2    | N/A           | Teacher submit certificates for verification | Teacher → Verifier | Verifier approve/reject |

|   |     |                                         |                           |                                 |
|---|-----|-----------------------------------------|---------------------------|---------------------------------|
| 3 | N/A | Teacher tạo class (sau khi verified)    | Teacher → System          | Set price, duration, schedule   |
| 4 | N/A | Teacher nhận booking notification       | Teacher → System          | Auto-accept hoặc manual accept? |
| 5 | N/A | Teacher conduct video class             | Teacher → Jitsi           | Record attendance?              |
| 6 | N/A | Teacher receive credit (sau commission) | System → Wallet           | 90% của class price             |
| 7 | N/A | Teacher withdraw credit                 | Teacher → Payment Gateway | Withdrawal process TBD          |

## 5.3 Verification Flow

| Step | Current state | Future state                       | Actor/system      | Rule or exception             |
|------|---------------|------------------------------------|-------------------|-------------------------------|
| 1    | N/A           | Student submit verification docs   | User → System     | Student ID / enrollment photo |
| 2    | N/A           | Admin review student docs          | Admin → System    | Manual review                 |
| 3    | N/A           | Teacher submit skill certificates  | Teacher → System  | Upload documents              |
| 4    | N/A           | Verifier review certificates       | Verifier → System | Domain expert review          |
| 5    | N/A           | Notification of approval/rejection | System → User     | Reason provided               |

## 6. Requirements

### Business Requirements

| ID     | Type     | Requirement                                                                            | Priority | Source | Status |
|--------|----------|----------------------------------------------------------------------------------------|----------|--------|--------|
| BR-001 | Business | Hệ thống phải cho phép học viên nạp tiền thật và chuyển thành credit nội bộ            | Must     | Intent | Draft  |
| BR-002 | Business | Hệ thống phải debit credit từ học viên khi book và credit cho giáo viên khi hoàn thành | Must     | Intent | Draft  |
| BR-003 | Business | Hệ thống phải thu 10% commission trên mỗi giao dịch class                              | Must     | Intent | Draft  |

|        |          |                                                            |        |        |       |
|--------|----------|------------------------------------------------------------|--------|--------|-------|
| BR-004 | Business | Giáo viên phải được verify trước khi có thể tạo lớp        | Must   | Intent | Draft |
| BR-005 | Business | Học viên phải verify student ID trước khi sử dụng platform | Must   | Intent | Draft |
| BR-006 | Business | Hệ thống phải hỗ trợ video call cho online classes         | Must   | Intent | Draft |
| BR-007 | Business | Low-rated profiles phải bị giảm visibility trong search    | Should | Intent | Draft |

## Functional Requirements

| ID     | Type       | Requirement                                                                                | Priority | Source | Status |
|--------|------------|--------------------------------------------------------------------------------------------|----------|--------|--------|
| FR-001 | Functional | User có thể đăng ký tài khoản với email/password                                           | Must     | Intent | Draft  |
| FR-002 | Functional | User có thể upload student ID/enrollment confirmation để verify                            | Must     | Intent | Draft  |
| FR-003 | Functional | Admin có thể approve/reject student verification                                           | Must     | Intent | Draft  |
| FR-004 | Functional | Teacher có thể submit certificates/proofs để xin verify skill                              | Must     | Intent | Draft  |
| FR-005 | Functional | Verifier có thể approve/reject teacher skill verification                                  | Must     | Intent | Draft  |
| FR-006 | Functional | Verified teacher có thể tạo class với title, description, skill, price, duration, schedule | Must     | Intent | Draft  |
| FR-007 | Functional | Learner có thể browse và search classes theo skill, price, rating                          | Must     | Intent | Draft  |
| FR-008 | Functional | Learner có thể book class nếu đủ credit và book 24h trước                                  | Must     | Intent | Draft  |
| FR-009 | Functional | Hệ thống tự động debit credit khi book, credit cho teacher khi complete                    | Must     | Intent | Draft  |
| FR-010 | Functional | User có thể nạp tiền qua payment gateway                                                   | Must     | Intent | Draft  |
| FR-011 | Functional | User có thể xem transaction history                                                        | Must     | Intent | Draft  |

|        |            |                                                           |        |        |       |
|--------|------------|-----------------------------------------------------------|--------|--------|-------|
| FR-012 | Functional | User có thể withdraw credit                               | Must   | Intent | Draft |
| FR-013 | Functional | Class link (Jitsi) được gửi cho cả learner và teacher     | Must   | Intent | Draft |
| FR-014 | Functional | Learner có thể đánh giá teacher sau class                 | Must   | Intent | Draft |
| FR-015 | Functional | Hệ thống có in-platform messaging giữa learner và teacher | Must   | Intent | Draft |
| FR-016 | Functional | Class duration phải từ 30 phút đến 3 giờ                  | Must   | Intent | Draft |
| FR-017 | Functional | Profile visibility giảm khi rating thấp                   | Should | Intent | Draft |

## 7. Use Cases and User Stories

### UC-001: Learner Registration and Verification

- **Actor:** Learner
- **Trigger:** User clicks "Register"
- **Preconditions:** User chưa có tài khoản
- **Main flow:**
  1. User nhập email, password, full name
  2. User upload photo của student ID hoặc enrollment confirmation
  3. System lưu documents và gửi notification cho admin
  4. Admin review và approve/reject
  5. System notify user về kết quả
- **Alternate/error flows:**
  - Reject: User nhận được lý do, có thể resubmit
  - Timeout: Reminder cho admin sau X hours
- **Postconditions:** User có verified student status hoặc rejection reason

### UC-002: Teacher Verification

- **Actor:** Teacher, Verifier
- **Trigger:** Verified student muốn trở thành teacher
- **Preconditions:** User đã verified student status
- **Main flow:**

1. Teacher submit certificates/proofs của skill
2. System assign to verifiers có domain expertise
3. Verifier review và approve/reject
4. System notify teacher

- **Alternate/error flows:**

- Reject: Teacher nhận feedback, có thể resubmit
- Escalation: Complex cases có thể escalate

- **Postconditions:** Teacher có verified teacher status

## UC-003: Class Creation

- **Actor:** Teacher

- **Trigger:** Verified teacher clicks "Create Class"

- **Preconditions:** Teacher đã verified

- **Main flow:**

1. Teacher nhập title, description
2. Teacher chọn skill category
3. Teacher set price (bằng credit)
4. Teacher set duration (30min-3h)
5. Teacher set schedule (phải >24h từ now)
6. Teacher save class

- **Postconditions:** Class visible to learners

## UC-004: Class Booking

- **Actor:** Learner

- **Trigger:** Learner clicks "Book" on a class

- **Preconditions:** Learner verified, đủ credit, class available

- **Main flow:**

1. Learner xem class details
2. Learner click "Book"
3. System check: sufficient credit, 24h advance booking rule
4. System debit learner's wallet
5. System create booking record
6. System notify teacher
7. System send confirmation to learner

- **Alternate/error flows:**

- Insufficient credit: Error message, prompt to top-up

- Less than 24h: Error message
- Teacher cancels: Refund (policy TBD)
- **Postconditions:** Booking confirmed, credit debited, teacher notified

## UC-005: Class Execution

- **Actor:** Teacher, Learner
- **Trigger:** Đến scheduled time
- **Preconditions:** Booking confirmed
- **Main flow:**
  1. System gửi Jitsi link cho cả teacher và learner (15 min before)
  2. Teacher và learner join video call
  3. Class diễn ra
  4. Teacher hoặc learner end call
  5. System mark class as completed
  6. System credit teacher wallet (90% of price)
  7. System credit platform commission (10%)
- **Alternate/error flows:**
  - Technical issues: Retry connection
  - No-show: Policy TBD (affects rating)
- **Postconditions:** Class completed, teacher credited

## UC-006: Rating and Review

- **Actor:** Learner
- **Trigger:** Class completed
- **Preconditions:** Class đã completed
- **Main flow:**
  1. System prompt learner để rate (1-5) và comment
  2. Learner submit rating
  3. System update teacher's average rating
  4. If rating below threshold, teacher's visibility reduced
- **Postconditions:** Rating recorded, teacher's profile updated

## US-001: As a learner, I want to top up my wallet so that I can book classes

Acceptance criteria

- Given I am logged in as a verified learner, when I navigate to wallet and enter amount, then I should see payment gateway options.
- Given I have sufficient balance, when payment completes, then my credit balance should increase by the equivalent amount (1 credit = 1,000 VND).
- Given a failed payment, when payment gateway returns error, then my credit balance should remain unchanged and I should see an error message.

## US-002: As a teacher, I want to set my own class price so that I can earn appropriately for my expertise

### Acceptance criteria

- Given I am a verified teacher, when I create a class, then I should be able to set the price in credits.
- Given I set a price, when a learner books my class, then I should receive 90% of the price after commission.

## US-003: As a learner, I want to book a class at least 24 hours in advance so that I can prepare

### Acceptance criteria

- Given I am viewing a class, when I try to book with less than 24 hours notice, then I should see an error message.
- Given I am viewing a class with enough credit, when I book more than 24 hours in advance, then my booking should be confirmed and credit debited.

## 8. Data Model

### 8.1 Core Entities

| Entity              | Key fields                                                                                   | Relationships                      | Lifecycle                   | Classification | Owner  |
|---------------------|----------------------------------------------------------------------------------------------|------------------------------------|-----------------------------|----------------|--------|
| User                | id, email, password_hash, full_name, role, verification_status, created_at                   | Has wallet, can be learner/teacher | Active, Suspended, Deleted  | PII            | System |
| StudentVerification | id, user_id, document_type, document_url, status, reviewed_by, reviewed_at, rejection_reason | Belongs to User                    | Pending, Approved, Rejected | PII            | Admin  |

|                     |                                                                                                             |                                  |                                        |           |          |
|---------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------|-----------|----------|
| TeacherVerification | id, user_id, certificates[], skill_area, status, verified_by, verified_at, rejection_reason                 | Belongs to User                  | Pending, Approved, Rejected            | PII       | Verifier |
| Skill               | id, name, parent_id, is_active                                                                              | Has many Classes                 | Active, Inactive                       | Public    | Admin    |
| Class               | id, teacher_id, skill_id, title, description, price_credits, duration_minutes, schedule, status, created_at | Belongs to Teacher, has Bookings | Draft, Published, Cancelled, Completed | Public    | Teacher  |
| Booking             | id, class_id, learner_id, booked_at, status, credit_amount, commission_amount                               | Belongs to Class, Learner        | Confirmed, Cancelled, Completed        | Public    | System   |
| Transaction         | id, wallet_id, type, amount, balance_after, reference_type, reference_id, created_at                        | Belongs to Wallet                | Completed, Failed                      | Public    | System   |
| Wallet              | id, user_id, balance_credits, created_at, updated_at                                                        | Belongs to User                  | Active                                 | Financial | System   |
| Review              | id, booking_id, rating, comment, created_at                                                                 | Belongs to Booking               | Created                                | Public    | Learner  |
| Message             | id, sender_id, receiver_id, class_id, content, created_at, read_at                                          | Between Users                    | Sent, Read                             | PII       | System   |

## 8.2 Data Rules

- Wallet balance cannot be negative
- Booking only allowed when wallet balance  $\geq$  class price
- Teacher can only create class when verification status = Approved
- Class schedule must be at least 24 hours from current time
- Class duration must be between 30 and 180 minutes

## 9. API and Integration Contract

### 9.1 Internal APIs

| API/Event ID | Method/path                | Purpose                     | Auth                    | Request/response                                          | Errors   |
|--------------|----------------------------|-----------------------------|-------------------------|-----------------------------------------------------------|----------|
| API-001      | POST /auth/register        | User registration           | None                    | {email, password, full_name}                              | 400, 409 |
| API-002      | POST /auth/login           | User login                  | None                    | {email, password}                                         | 401      |
| API-003      | POST /users/verify-student | Submit student verification | User                    | {document_type, document_url}                             | 400, 403 |
| API-004      | GET /users/me              | Get current user profile    | JWT                     | -                                                         | 401      |
| API-005      | POST /teachers/verify      | Submit teacher verification | User (verified student) | {certificates[], skill_area}                              | 400, 403 |
| API-006      | GET /skills                | List all skill categories   | Public                  | -                                                         | -        |
| API-007      | POST /classes              | Create new class            | JWT (verified teacher)  | {title, description, skill_id, price, duration, schedule} | 400, 403 |
| API-008      | GET /classes               | List/search classes         | Public                  | Query: skill_id, min_price, max_price, min_rating         | -        |
| API-009      | GET /classes/{id}          | Get class details           | Public                  | -                                                         | 404      |

|         |                               |                           |                             |                                  |                     |
|---------|-------------------------------|---------------------------|-----------------------------|----------------------------------|---------------------|
| API-010 | POST<br>/classes/{id}/book    | Book a class              | JWT<br>(verified learner)   | -                                | 400,<br>402,<br>404 |
| API-011 | POST<br>/classes/{id}/cancel  | Cancel a class/booking    | JWT<br>(teacher or learner) | -                                | 400,<br>403         |
| API-012 | GET /bookings                 | List user's bookings      | JWT                         | Query: status                    | -                   |
| API-013 | POST /wallet/topup            | Top up wallet             | JWT                         | {amount, payment_method}         | 400                 |
| API-014 | GET /wallet                   | Get wallet balance        | JWT                         | -                                | -                   |
| API-015 | POST /wallet/withdraw         | Withdraw to real money    | JWT                         | {amount}                         | 400,<br>402         |
| API-016 | GET /wallet/transactions      | Get transaction history   | JWT                         | Query: page, limit               | -                   |
| API-017 | POST /reviews                 | Submit review             | JWT                         | {booking_id, rating, comment}    | 400,<br>403         |
| API-018 | GET /classes/{id}/reviews     | Get class/teacher reviews | Public                      | -                                | -                   |
| API-019 | GET /messages                 | Get messages              | JWT                         | Query: class_id, other_user_id   | -                   |
| API-020 | POST /messages                | Send message              | JWT                         | {receiver_id, class_id, content} | 400,<br>403         |
| API-021 | GET /classes/{id}/jitsi-token | Get Jitsi meeting token   | JWT                         | -                                | 403,<br>404         |

## 9.2 External Integrations

| Integration     | Purpose                            | Out of Scope               | Notes |
|-----------------|------------------------------------|----------------------------|-------|
| Payment Gateway | Real-money top-ups and withdrawals | Chọn provider ở spec stage | TBD   |

|              |                                 |   |                            |
|--------------|---------------------------------|---|----------------------------|
| Jitsi        | Video call for classes          | - | Self-hosted or meet.jit.si |
| File Storage | Store uploaded documents/images | - | S3, Cloudinary, etc.       |

## 10. Non-Functional Requirements

| ID      | Category      | Target                                | Measurement           | Priority | Owner      |
|---------|---------------|---------------------------------------|-----------------------|----------|------------|
| NFR-001 | Performance   | Class search response < 500ms         | p95 latency           | Must     | Backend    |
| NFR-002 | Performance   | Booking confirmation < 2s             | End-to-end            | Must     | Backend    |
| NFR-003 | Availability  | 99.5% uptime                          | Monthly SLA           | Must     | DevOps     |
| NFR-004 | Scalability   | Support 1000 concurrent users for MVP | Load test             | Should   | Backend    |
| NFR-005 | Security      | All APIs over HTTPS                   | TLS 1.2+              | Must     | DevOps     |
| NFR-006 | Security      | JWT tokens expire after 24h           | Token validation      | Must     | Backend    |
| NFR-007 | Security      | Wallet operations are atomic          | Transaction integrity | Must     | Backend    |
| NFR-008 | Privacy       | PII encrypted at rest                 | Encryption validation | Must     | DevOps     |
| NFR-009 | Privacy       | Data retention policy documented      | Policy doc            | Must     | Compliance |
| NFR-010 | Accessibility | WCAG 2.1 AA for web                   | Audit                 | Should   | Frontend   |
| NFR-011 | Mobile        | Responsive design, usable on 320px+   | UX testing            | Must     | Frontend   |
| NFR-012 | Operations    | Logging for all wallet transactions   | Log retention         | Must     | Backend    |

|         |            |                                    |                   |        |        |
|---------|------------|------------------------------------|-------------------|--------|--------|
| NFR-013 | Operations | Error tracking (Sentry equivalent) | Incident response | Should | DevOps |
|---------|------------|------------------------------------|-------------------|--------|--------|

## 11. Security, Privacy and Compliance

- **Authentication and authorization:**
  - JWT-based authentication
  - Role-based access: Learner, Teacher, Verifier, Admin
  - Teacher-only endpoints require teacher verification status
  - All wallet operations require authentication
- **Sensitive data and classification:**
  - PII: email, full\_name, student documents, messages → encrypted at rest
  - Financial: wallet balance, transactions → audited
  - Public: class listings, reviews, skill categories
- **Audit and traceability:**
  - All wallet transactions logged with reference to booking/payment
  - Verification decisions logged with verifier ID and timestamp
  - No personal info in logs (use user\_id)
- **Retention/deletion:**
  - User accounts: soft delete, data retained per policy
  - Transactions: retained for financial compliance
  - Documents: deleted when account deleted
- **Applicable policy or regulation:**
  - Vietnamese payment regulations (if applicable)
  - PDPL (Personal Data Protection Law) compliance

## 12. Delivery Plan and Dependencies

| Increment | Scope                                            | Dependency             | Exit criteria                              | Risk                           |
|-----------|--------------------------------------------------|------------------------|--------------------------------------------|--------------------------------|
| 1         | User auth + student verification + wallet top-up | Payment gateway (mock) | User can register, verify, top-up          | Payment integration complexity |
| 2         | Teacher verification + class creation            | Increment 1            | Teacher can submit certs, admin can verify | Verifier workflow              |

|   |                                            |             |                                   |                        |
|---|--------------------------------------------|-------------|-----------------------------------|------------------------|
| 3 | Class search + booking + Jitsi integration | Increment 2 | Learner can book, join video call | Jitsi integration      |
| 4 | Rating + messaging + withdrawals           | Increment 3 | Full flow complete                | Withdrawal process TBD |

## 13. Traceability Matrix

| Business goal                  | Requirement            | Use case/story | API/data/NFR                              | Test evidence               |
|--------------------------------|------------------------|----------------|-------------------------------------------|-----------------------------|
| Learner nạp tiền               | BR-001, FR-010, FR-011 | US-001         | API-013, API-014, API-016, Wallet entity  | Wallet top-up test          |
| Book class có debit/credit     | BR-002, FR-008, FR-009 | UC-004, UC-005 | API-010, API-012, Booking entity, NFR-007 | Booking flow test           |
| Commission 10%                 | BR-003                 | UC-004, UC-005 | Transaction entity, NFR-012               | Commission calculation test |
| Teacher verified trước khi dạy | BR-004, FR-004, FR-005 | UC-002         | API-005, TeacherVerification entity       | Verification workflow test  |
| Student verified               | BR-005, FR-002, FR-003 | UC-001         | API-003, StudentVerification entity       | Student verification test   |
| Video call                     | BR-006, FR-013         | UC-005         | API-021, Jitsi integration, NFR-001       | Video call E2E test         |
| Rating affects visibility      | BR-007, FR-014, FR-017 | UC-006         | API-017, API-018, Review entity           | Rating impact test          |

## 14. Risks and Decisions

| ID    | Risk/decision                                 | Impact | Likelihood | Owner      | Mitigation/status                                                |
|-------|-----------------------------------------------|--------|------------|------------|------------------------------------------------------------------|
| R-001 | Fraudulent top-ups/withdrawals                | High   | Medium     | Security   | Payment gateway fraud detection, manual review for large amounts |
| R-002 | Verifier collusion                            | High   | Low        | Operations | Rotate verifiers, audit logs, escalation process                 |
| R-003 | Dispute resolution (no formal process in MVP) | Medium | Medium     | Product    | Document escalation path, collect feedback for v2                |

|       |                               |          |        |            |                                                                 |
|-------|-------------------------------|----------|--------|------------|-----------------------------------------------------------------|
| R-004 | Fake documents passing review | Medium   | Medium | Operations | Better training for verifiers, periodic re-verification         |
| R-005 | Wallet ledger bugs            | Critical | Low    | Backend    | Atomic transactions, double-entry accounting, extensive testing |
| R-006 | Low teacher supply at launch  | Medium   | High   | Product    | Partner with student organizations, incentive programs          |

## 15. Quality Review

- **Quality gate:** Not started
- **Unresolved questions:**
  1. Skill category structure (hierarchical vs flat)?
  2. Rating threshold để giảm visibility?
  3. Refund policy cho cancelled classes?
  4. Payment gateway provider?
  5. Withdrawal process flow?
- **Reviewers:** Group 4
- **Approval decision:** Pending